

Complete Mathematical & Physical Formulary

Dark Vessel Detection via Multimodal SAR & Context-Guided Cross-Attention (CGCAM)

Course: Computer Vision (A.Y. 2025–2026) — Sapienza University of Rome

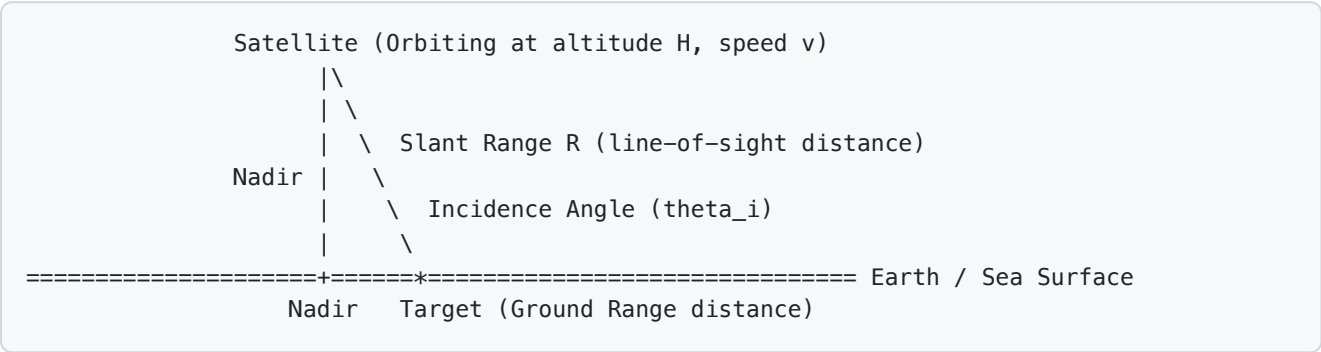
Associated Document: [PROJECT_PLAN.md](#)

Target Audience: Computer Science & Artificial Intelligence Students

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1. Radar & SAR Physical Geometry



1.1 Radar Range Equation (Slant Range)

$$R = \frac{c \cdot \Delta t}{2}$$

- **Conceptual Meaning:** Calculates the straight-line Euclidean distance (*Slant Range*) between the satellite antenna and a target on Earth by measuring the round-trip flight time of a microwave pulse.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
R	Slant Range Distance	$\mathbb{R}^+$	The line-of-sight distance from satellite to target (meters [m]).
c	Speed of Light	Constant	Universal speed of electromagnetic wave propagation in vacuum/air ($\approx 3 \cdot 10^8$ m/s).
Δt	Two-Way Travel Time	$\mathbb{R}^+$	Time elapsed between pulse transmission and echo reception (seconds [s]).
2	Round-Trip Path Factor	Constant	Compensates for the fact that the wave travels to the target and back.

1.2 Real Aperture Radar (RAR) Azimuth Resolution

$$\delta_{az,RAR} = R \cdot \theta_{az} \approx \frac{\lambda R}{L}$$

- **Conceptual Meaning:** In a conventional stationary antenna (*Real Aperture Radar*), optical diffraction spreads the microwave beam as it travels. Consequently, spatial resolution along the flight track (*Azimuth*) degrades linearly with distance.

- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$\delta_{az,RAR}$	RAR Azimuth Resolution	$\mathbb{R}^+$	Minimum distance along the flight path between two distinguishable targets ([m]).
R	Slant Range Distance	$\mathbb{R}^+$	Range from radar to ground target ([m]).
θ_{az}	Azimuth Beamwidth	$\mathbb{R}^+$	Angular angular spread of the radiated microwave beam (radians [rad]).
λ	Radar Wavelength	$\mathbb{R}^+$	Carrier wavelength (meters [m], ≈ 0.0554 m for Sentinel-1 C-band).
L	Antenna Physical Length	$\mathbb{R}^+$	Physical aperture length of the antenna mounted on the satellite ([m]).

1.3 Synthetic Aperture Radar (SAR) Azimuth Resolution

$$\delta_{az,SAR} = \frac{L}{2}$$

- **Conceptual Meaning:** As the satellite moves, it illuminates a ground target from multiple successive orbital positions, recording both amplitude and Doppler phase. A matched filter reconstructs an equivalent *synthetic antenna* kilometers long. Incredibly, the resulting spatial resolution becomes **independent of orbital altitude R and wavelength λ** , depending solely on antenna length.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$\delta_{az,SAR}$	SAR Azimuth Resolution	$\mathbb{R}^+$	The theoretical along-track resolution limit ([m]).
L	Physical Antenna Length	$\mathbb{R}^+$	Real length of the antenna ([m]). For Sentinel-1, $L \approx 12.3$ m $\implies \delta_{az} \approx 6.1$ m.
2	Two-Way Doppler Factor	Constant	Arises from the two-way phase history during coherent synthesis.

1.4 Slant Range Resolution

$$\delta_r = \frac{c}{2B}$$

- **Conceptual Meaning:** Defines how well the radar can separate two targets along the oblique line-of-sight direction. Instead of using infinitesimally short pulses (which lack energy), SAR transmits frequency-modulated *chirps* of bandwidth B , achieving fine time resolution through pulse compression.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
δ_r	Slant Range Resolution	$\mathbb{R}^+$	Minimum resolvable distance along the radar beam axis ([m]).
c	Speed of Light	Constant	$\approx 3 \cdot 10^8$ m/s.
B	Chirp Bandwidth	$\mathbb{R}^+$	Frequency excursion bandwidth of the transmitted radar pulse (Hertz [Hz]).
2	Round-Trip Factor	Constant	Accounts for the wave travelling to and from the target.

1.5 Ground Range Resolution

$$\delta_{gr} = \frac{\delta_r}{\sin \theta_i} = \frac{c}{2B \sin \theta_i}$$

- **Conceptual Meaning:** The slant range resolution δ_r is measured obliquely in space. To know the true resolution projected horizontally onto the sea or land surface (*Ground Range*), we project δ_r using trigonometry through the local incidence angle.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
δ_{gr}	Ground Range Resolution	$\mathbb{R}^+$	Spatial resolution projected on the horizontal Earth surface ([m]).
δ_r	Slant Range Resolution	$\mathbb{R}^+$	Radial resolution along the beam line ([m]).
θ_i	Local Incidence Angle	$(0, \frac{\pi}{2})$	Angle between the incoming radar ray and the local surface normal vector.
$\sin \theta_i$	Projection Factor	$(0, 1)$	Trigonometric scaling factor projecting slant range onto horizontal ground range.

2. Signal Representation & Radiometry

2.1 Complex SAR Signal (Single Look Complex - SLC)

$$S(x, y) = I(x, y) + j \cdot Q(x, y) = A(x, y) \cdot e^{j\phi(x, y)}$$

- Conceptual Meaning:** Before amplitude detection, SAR data is fundamentally complex-valued. Each pixel stores two orthogonal components of the returned electromagnetic wave: amplitude (energy backscattered) and coherent phase (precise millimeter wave travel distance).
- Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$S(x, y)$	Complex Radar Signal	$\mathbb{C}$	Complex value at image coordinate pixel (x, y) .
$I(x, y)$	In-Phase Component	$\mathbb{R}$	Real part of the received signal vector.
$Q(x, y)$	Quadrature Component	$\mathbb{R}$	Imaginary part (shifted by 90°).
j	Imaginary Unit	Constant	$j = \sqrt{-1}$.
$A(x, y)$	Signal Amplitude	$\mathbb{R}^+$	Wave magnitude: $A = \sqrt{I^2 + Q^2}$. Proportional to reflected power.
$\phi(x, y)$	Coherent Phase	$[-\pi, \pi)$	Phase angle: $\phi = \text{atan2}(Q, I)$. Sensitive to terrain height/displacement.

2.2 Normalized Radar Cross Section (Sigma Nought, σ^0)

$$\sigma^0 = \frac{\langle \sigma \rangle}{A_{\text{ground}}}$$

- Conceptual Meaning:** Measures the electromagnetic reflectivity of a distributed continuous surface (like the ocean). It normalizes the average radar cross section of all micro-scatterers by the physical ground area of the resolution cell.
- Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
σ^0	Sigma Nought (Backscatter)	$\mathbb{R}^+$	Dimensionless normalized backscattering coefficient ($[\text{m}^2/\text{m}^2]$).
$\langle \sigma \rangle$	Mean Radar Cross Section	$\mathbb{R}^+$	Expected radar cross-sectional reflective area of the target ($[\text{m}^2]$).
A_{ground}	Physical Ground Area	$\mathbb{R}^+$	Footprint area of a single resolution cell projected onto the ground ($[\text{m}^2]$).

2.3 Decibel (dB) Conversion

$$\sigma^0_{[\text{dB}]} = 10 \cdot \log_{10}(\sigma^0_{\text{linear}})$$

- Conceptual Meaning:** Radar backscatter spans several orders of magnitude (from 0.0001 for flat calm sea to > 1000 for metallic superstructures). The decibel scale compresses this exponential distribution into a manageable linear-like scale for analysis and visualization.
- Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$\sigma^0_{[\text{dB}]}$	Backscatter in Decibels	$\mathbb{R}$	Logarithmic radiometric intensity ($[\text{dB}]$). Typical range: -35 dB to $+15 \text{ dB}$.
σ^0_{linear}	Linear Power Ratio	$\mathbb{R}^+$	Raw power ratio ($[\text{m}^2/\text{m}^2]$).
$10 \log_{10}$	Decibel Operator	Function	Standard acoustic/electromagnetic power scaling transformation.

2.4 Multiplicative Speckle Noise Model

$$I(x, y) = R(x, y) \cdot \eta(x, y)$$

- Conceptual Meaning:** Unlike optical noise which is typically additive Gaussian ($I = R + \mathcal{N}$), SAR speckle is fundamentally **multiplicative**. It is caused by the constructive and destructive interference of coherent waves bouncing off sub-pixel microscopic scatterers. As a result, brighter surfaces exhibit larger absolute noise fluctuations.
- Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$I(x, y)$	Observed Pixel Intensity	$\mathbb{R}^+$	Raw corrupted intensity value observed by the detector.
$R(x, y)$	Ground Truth Reflectivity	$\mathbb{R}^+$	True underlying radar cross section to be estimated.
$\eta(x, y)$	Speckle Noise Factor	$\mathbb{R}^+$	Stationary stochastic noise process distributed as $\eta \sim \Gamma(L, 1/L)$.
L	Number of Looks	$\mathbb{N}^+$	Number of independent spatial looks averaged during image generation.
$\mathbb{E}[\eta]$	Expected Noise Mean	Constant	Equal to 1.0 (unbiased on average).
$\text{Var}[\eta]$	Noise Variance	$\mathbb{R}^+$	Equal to $1/L$. Decreases as more independent looks are multilooked.

2.5 Bragg Scattering Resonance Wavelength

$$\Lambda_{\text{sea}} = \frac{\lambda}{2 \sin \theta_i}$$

- **Conceptual Meaning:** Explains why rough seas look bright in VV polarization. When wind creates periodic surface ripples whose physical crest-to-crest spacing Λ_{sea} matches this equation, consecutive radar waves reflect in phase, constructively reinforcing the returned echo.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
Λ_{sea}	Ocean Wave Ripple Spacing	$\mathbb{R}^+$	Physical wavelength of the ocean capillary/gravity waves causing resonance ([m]).
λ	Radar Carrier Wavelength	$\mathbb{R}^+$	$\approx 0.0554 \text{ m}$ for Sentinel-1 C-band.
θ_i	Radar Incidence Angle	$(0, \frac{\pi}{2})$	Angle between the radar beam and the ocean surface normal.

2.6 Target-to-Clutter Ratio (TCR)

$$\text{TCR} = \frac{\sigma_{\text{target}}^0}{\sigma_{\text{clutter}}^0}$$

- Conceptual Meaning:** Evaluates how prominently a ship stands out against the ambient ocean background. In VV , sea clutter is high, so TCR is low. In VH , sea clutter is nearly zero while ship dihedral reflections remain strong, making TCR_{VH} immensely higher.
- Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
TCR	Target-to-Clutter Ratio	$\mathbb{R}^+$	Dimensionless contrast ratio (often expressed in dB as $\sigma^0_{\text{target, dB}} - \sigma^0_{\text{clutter, dB}}$).
σ^0_{target}	Target Backscatter	$\mathbb{R}^+$	Backscatter returned by the metallic vessel hull ($[\text{m}^2/\text{m}^2]$).
$\sigma^0_{\text{clutter}}$	Clutter Backscatter	$\mathbb{R}^+$	Ambient backscatter returned by the surrounding sea surface ($[\text{m}^2/\text{m}^2]$).

3. Geospatial & Radiometric Normalization

3.1 Radiometric Min-Max Standardization (dB Scale to [0, 1])

$$\sigma^0_{[\text{norm}]} = \text{clip}\left(\frac{\sigma^0_{[\text{dB}]} - \sigma^0_{\min}}{\sigma^0_{\max} - \sigma^0_{\min}}, 0.0, 1.0\right)$$

- Conceptual Meaning:** Linearly standardizes raw radar decibel values into the bounded range $[0.0, 1.0]$ required for stable neural network training (preventing gradient explosion or early activation saturation). Values outside the realistic physical range are clamped.
- Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$\sigma^0_{[\text{norm}]}$	Normalized Radar Pixel	$[0.0, 1.0]$	Dimensionless input feature value passed into the first convolution layer.
$\sigma^0_{[\text{dB}]}$	Input Backscatter	$\mathbb{R}$	Original decibel value stored in the GeoTIFF file.
$\sigma^0_{\min}$	Minimum Clipping Floor	Constant	Set to -35.0 dB (any lower value is sensor thermal noise).
$\sigma^0_{\max}$	Maximum Clipping Ceiling	Constant	Set to $+10.0$ dB (strong specular/corner metallic reflection).
$\text{clip}(x, a, b)$	Clamping Function	Function	$\text{clip}(x, a, b) = \max(a, \min(x, b))$.

3.2 Distance-to-Shore Normalization

$$\text{DistShore}_{[\text{norm}]} = \text{clip} \left(\frac{D}{D_{\text{max}}}, 0.0, 1.0 \right)$$

- **Conceptual Meaning:** Scales metric distance from the coast into $[0, 1]$. Distances beyond $D_{\text{max}} = 50 \text{ km}$ saturate to 1.0, preventing deep ocean areas from dominating feature gradients, while focusing high spatial sensitivity along the critical coastal margin where false alarms occur.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$\text{DistShore}_{[\text{norm}]}$	Normalized Shore Feature	$[0.0, 1.0]$	0.0 = Coastline, 1.0 = Deep Ocean $\geq 50 \text{ km}$.
D	Real Geodetic Distance	$\mathbb{R}^+$	Metric distance from current pixel to closest shoreline ([m]).
D_{max}	Normalization Horizon	Constant	Fixed at 50000.0 m (50 km).

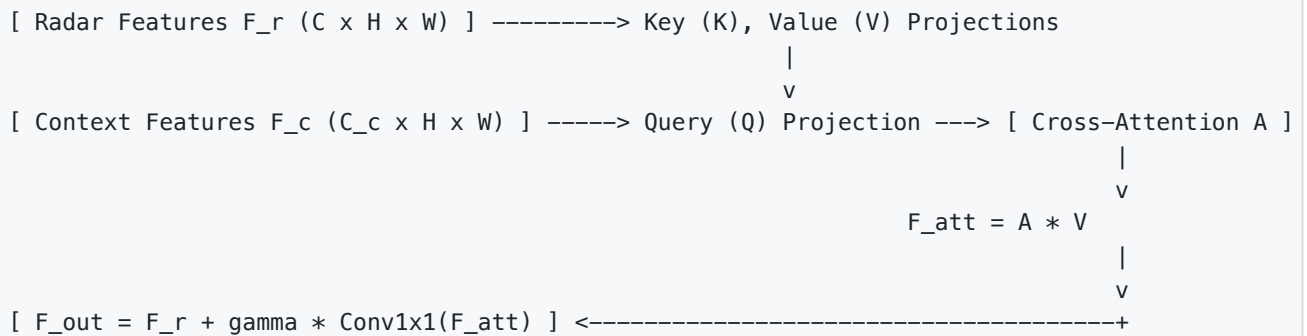
3.3 Ocean Bathymetry Normalization

$$\text{Bathy}_{[\text{norm}]} = \text{clip} \left(\frac{\text{depth}}{\text{depth}_{\text{max}}}, 0.0, 1.0 \right)$$

- **Conceptual Meaning:** Compresses seafloor depth into $[0, 1]$. Depths below $\text{depth}_{\text{max}} = -2000 \text{ m}$ saturate to 1.0. This ensures the model is sensitive to shallow waters, coral reefs, and continental shelves (0 to -200 m) where ships fish and navigate.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Physical Unit
$\text{Bathy}_{[\text{norm}]}$	Normalized Depth Feature	$[0.0, 1.0]$	0.0 = Sea Level (0 m), 1.0 = Abyssal Depth $\leq -2000 \text{ m}$.
depth	Ocean Floor Depth	$\mathbb{R}^-$	Negative metric bathymetry ([m]).
$\text{depth}_{\text{max}}$	Depth Clipping Threshold	Constant	Fixed at -2000.0 m .

4. Deep Learning Architecture & Novelty (CGCAM)



4.1 Linear Projections to Latent Space d

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{F}_c, \quad \mathbf{K} = \mathbf{W}_K \mathbf{F}_r, \quad \mathbf{V} = \mathbf{W}_V \mathbf{F}_r$$

- **Conceptual Meaning:** Projects heterogenous radar features (C channels) and geophysical context features (C_c channels) into a shared intermediate latent subspace of dimension d so their spatial correlation can be computed via dot-products.
- **Terms & Parameters:**

Symbol	Name	Shape / Domain	Definition & Conceptual Role
$\mathbf{F}_r$	Radar Backbone Features	$\mathbb{R}^{C \times H \times W}$	Latent representation extracted from VV, VH radar channels.
$\mathbf{F}_c$	Context Encoder Features	$\mathbb{R}^{C_c \times H \times W}$	Latent representation extracted from bathymetry & distance-to-shore.
$\mathbf{W}_Q$	Query Projection Weights	$\mathbb{R}^{d \times C_c}$	1×1 convolution weights projecting context features into Queries.
$\mathbf{W}_K$	Key Projection Weights	$\mathbb{R}^{d \times C}$	1×1 convolution weights projecting radar features into Keys.
$\mathbf{W}_V$	Value Projection Weights	$\mathbb{R}^{d \times C}$	1×1 convolution weights projecting radar features into Values.
$\mathbf{Q}$	Query Tensor	$\mathbb{R}^{d \times H \times W}$	Query: <i>"Where does geography indicate navigable ocean far from cliffs?"</i> .
$\mathbf{K}$	Key Tensor	$\mathbb{R}^{d \times H \times W}$	Key: <i>"Where did the radar observe high-energy reflections?"</i> .
$\mathbf{V}$	Value Tensor	$\mathbb{R}^{d \times H \times W}$	Value: Raw radar feature content to be modulated and preserved.
d	Latent Projection Dimension	$\mathbb{N}^+$	Intermediate feature channel capacity (e.g., $d = 64$).
H, W	Spatial Resolution	$\mathbb{N}^+$	Height and width of the bottleneck feature map (e.g., 32×32).

4.2 Scaled Dot-Product Spatial Cross-Attention

$$\mathbf{A}(i, j) = \text{Softmax} \left(\frac{\mathbf{Q}(i) \mathbf{K}(j)^T}{\sqrt{d}} \right)$$

- **Conceptual Meaning:** Computes a dense pairwise spatial compatibility matrix. If a radar pixel j has bright intensity but context pixel i indicates dry land or steep shoreline cliffs, the resulting attention weight is driven towards zero, effectively filtering out coastal false alarms.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Conceptual Role
$\mathbf{A}(i, j)$	Attention Weight	$[0, 1]$	Normalized importance of radar position j relative to context query i .
$\mathbf{Q}(i)$	Query Vector at Pixel i	$\mathbb{R}^d$	Geophysical context vector at flattened spatial index $i \in \{1, \dots, H \cdot W\}$.
$\mathbf{K}(j)^T$	Transposed Key Vector at j	$\mathbb{R}^d$	Radar feature vector at flattened spatial index $j \in \{1, \dots, H \cdot W\}$.
$\sqrt{d}$	Softmax Scaling Factor	$\mathbb{R}^+$	Scaling factor preventing inner products from growing excessively large in high dimensions.
Softmax	Normalization Operator	Function	Ensures all weights along spatial dimension j sum to 1.0 ($\sum_j \mathbf{A}(i, j) = 1$).

4.3 Gated Residual Modulation

$$\mathbf{F}_{\text{att}} = \mathbf{A}\mathbf{V}, \quad \mathbf{F}_{\text{out}} = \mathbf{F}_r + \gamma \cdot \text{Conv}_{1\times 1}(\mathbf{F}_{\text{att}})$$

- Conceptual Meaning:** Multiplies attention weights by the radar value representation ($\mathbf{A}\mathbf{V}$) to produce context-filtered features $\mathbf{F}_{\text{att}}$, then injects them back into the original radar stream through a learnable residual gate γ .
- Terms & Parameters:**

Symbol	Name	Shape / Domain	Definition & Conceptual Role
$\mathbf{F}_{\text{att}}$	Attention-Weighted Features	$\mathbb{R}^{d \times H \times W}$	Radar features filtered and modulated by geographic priors.
$\mathbf{F}_{\text{out}}$	Final Refined Output Map	$\mathbb{R}^{C \times H \times W}$	The conditioned representation passed to the prediction heads.
$\mathbf{F}_r$	Original Radar Features	$\mathbb{R}^{C \times H \times W}$	Identity shortcut connection preserving fine vessel details.
γ	Learnable Residual Gate	$\mathbb{R}$	Scalar parameter initialized to 0.0. Allows the network to gradually learn how much context to incorporate.
$\text{Conv}_{1 \times 1}$	Channel Projection Layer	Function	Re-projects dimension d back to the original backbone channel count C .

5. Objective Function & Inference Optimization (Option 1: Unified 2-Channel CenterNet)

5.1 Multi-Channel Modified Focal Loss

$$\mathcal{L}_{\text{total}} = -\frac{1}{N_{\text{targets}}} \sum_{c \in \{0,1\}} \sum_{x=1}^W \sum_{y=1}^H \begin{cases} (1 - \hat{Y}_{c,xy})^\alpha \log(\hat{Y}_{c,xy}) & \text{if } Y_{c,xy} = 1 \\ (1 - Y_{c,xy})^\beta (\hat{Y}_{c,xy})^\alpha \log(1 - \hat{Y}_{c,xy}) & \text{otherwise} \end{cases}$$

- **Conceptual Meaning:** Solves extreme spatial class imbalance between millions of ocean background pixels and sparse vessel/structure centers across both semantic channels. It downweights easy ocean pixels by $(1 - \hat{Y}_{c,xy})^\alpha \approx 10^{-4}$ while dynamically preserving large gradients on misdetected targets.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Conceptual Role
$\mathcal{L}_{\text{total}}$	Total Objective Function	$\mathbb{R}^+$	Multi-channel focal loss minimized via AdamW.
c	Semantic Channel Index	$\{0, 1\}$	$c = 0$: Mobile Vessels, $c = 1$: Fixed Offshore Structures.
N_{targets}	Total Ground-Truth Targets	$\mathbb{N}^+$	Total annotated maritime objects in the batch ($N = \sum_c N_c$).
x, y	Grid Pixel Coordinates	$\mathbb{N} \times \mathbb{N}$	Spatial position on the 256×256 feature heatmap ($x \in [1, W], y \in [1, H]$).
$Y_{c,xy}$	Ground Truth Target Heatmap	$[0.0, 1.0]$	Continuous Gaussian peak centered on ground-truth target centroid.
$\hat{Y}_{c,xy}$	Predicted Heatmap Activation	$[0.0, 1.0]$	Model's sigmoid output probability at channel c and position (x, y) .
α	Modulating Exponent	Constant	Set to 2.0. Suppresses loss on easy, well-classified background points.
β	Gaussian Distance Exponent	Constant	Set to 4.0. Softens the loss for predictions slightly off-center ($Y_{c,xy} > 0$).

5.2 Ground-Truth Gaussian Target Generation

$$Y_{c,xy} = \max_{k \in \mathcal{T}_c} \exp \left(-\frac{(x-x_k)^2 + (y-y_k)^2}{2\sigma^2} \right)$$

- **Conceptual Meaning:** Renders target peaks as continuous 2D Gaussians rather than discrete binary impulses (1-hot), avoiding harsh penalties on predictions within 1–2 pixels of the true center. Overlapping targets take the pointwise maximum.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Conceptual Role
$\mathcal{T}_c$	Target Set for Class c	Set	Collection of annotated centroids $\{(x_k, y_k)\}$ belonging to class c .
(x_k, y_k)	Annotated Ground-Truth Centroid	$\mathbb{N} \times \mathbb{N}$	Exact coordinates of the k -th maritime target.
σ	Gaussian Standard Deviation	$\mathbb{R}^+$	Spatial radius set to $\sigma = 2.0$ pixels (representing ≈ 20 m at 10 m/pixel).

5.3 Anchor-Free Peak Extraction via Local Max-Pooling (3x3 NMS)

$$\hat{\mathcal{P}}_c = \left\{ (x, y) \mid \hat{Y}_{c,xy} = \max_{(u,v) \in \mathcal{N}_3(x,y)} \hat{Y}_{c,uv} \quad \text{and} \quad \hat{Y}_{c,xy} \geq \tau \right\}$$

- **Conceptual Meaning:** Efficient post-processing replacing complex bounding-box NMS. A 3×3 max-pooling operation identifies local maxima; pixels that match the max-pooled map and exceed the confidence threshold τ are extracted as detected instances.
- **Terms & Parameters:**

Symbol	Name	Domain	Definition & Conceptual Role
$\hat{\mathcal{P}}_c$	Set of Detected Detections	Set	Extracted coordinates of candidate objects for class c .
$\mathcal{N}_3(x, y)$	3×3 Pixel Neighborhood	Set	The 8-connected local neighborhood around pixel (x, y) .
τ	Confidence Threshold	$[0.0, 1.0]$	Minimum activation required (e.g., $\tau = 0.3$).

5.4 Metric Evaluation: Spatial Distance Matching & F1-Score

A predicted centroid $(\hat{x}, \hat{y}) \in \hat{\mathcal{P}}_c$ is matched as a **True Positive (TP)** to a ground-truth centroid (x_k, y_k) if: $d_{\text{ground}}((\hat{x}, \hat{y}), (x_k, y_k)) = \sqrt{(\hat{x} - x_k)^2 + (\hat{y} - y_k)^2} \cdot s_{\text{pixel}} \leq D_{\text{match}}$ where $s_{\text{pixel}} = 10$ m/pixel and $D_{\text{match}} = 100$ m (10 pixels).

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad F_{1,c} = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

- Conceptual Meaning:** Standard maritime evaluation metric for Sentinel-1 SAR. A detection is declared valid if it falls within 100 meters of the ground truth, computing harmonic mean F_1 separately for vessels ($c = 0$) and fixed structures ($c = 1$).

6. Academic Grading Formulation

6.1 Final Course Grade Composition

Final Grade = $0.5 \cdot G_{\text{written}} + 0.5 \cdot G_{\text{project}}$

- Conceptual Meaning:** Standard arithmetic weighting governing the Computer Vision master's exam at Sapienza University of Rome. Both parts must meet the sufficiency threshold ($G \geq 18$) for the final grade to be recorded.
- Terms & Parameters:**

Symbol	Name	Domain	Definition
Final Grade	Registered Exam Grade	[18, 30L]	Final grade recorded on Infostud.
G_{written}	Written Exam Grade	[18, 32]	Score obtained on the 1-hour written exam (4 questions $\times$ 8 points).
G_{project}	Project Assessment	[18, 30L]	Combined grade for code quality, novelty implementation, and oral defense.
0.5	Component Weight	Constant	Equal 50%/50% distribution.